Agenda

* A/B Testing & Launch Recommendations

Assignments

Made with Goodnotes

RFM → Monetary

Recency

Frequency

customer segmentation

Made with Goodnotes

4 different kinds of product sense problems

1. Product Diagnostics - Analyse the metric change
2. New product / feature - Measuring performance / Success
3. Product design - Launch feature Recommendation
4. Product Improvement

Made with Goodnotes

Product Diagnostics

C – clarify ✓

R – Ruleout ✓

I – Internal Data ✓

E – External Data ✓

D

T – Time ✓

R – Region ✓

O – Other related feature ✓

Pi – platform ✓

C – canibalization ✓

S – segmentation ✓

Design a metric to measure feature / product

- Clarify
- plan
- Summarize
  - NSM, L1, L2

Business plan ✓

Design a metric ✓ – Awareness ✓, Acquisition ✓,

Adoption ✓, Engagement ✓, Revenue ✓, Retention ✓, Gaurdrail metric ✓

Made with Goodnotes

Objective

* Business sense towards A/B testing

** How to launch A/B testing to get "valid" and "reliable" results

- Developing Hypothesis (NULL, Alternative)
- Designing A/B Tests
- Evaluating test results
- Making decisions

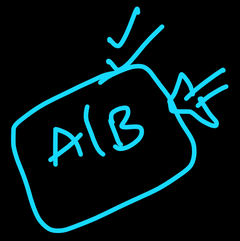
* Launch recommendations and pitfalls with A/B testing setups.

Made with Goodnotes

Examples

1. UI/UX changes

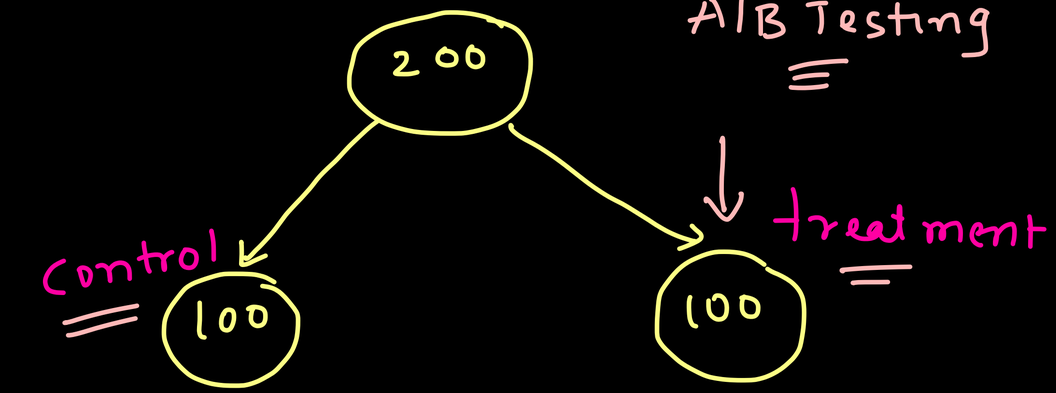
↳ Landing page optimization



1.

Covid → Remidiviser

XYZ



no

Judgment Criteria & General Framework

* Clarify

* Design an Experiment

  - Develop the right Hypothesis (H_0, H_a)
  - *** Sample size calculation
  - Duration
  - Pitfalls

* Identifying the right metric for testing significance - North star metric

Made with Goodnotes

Case Study

How would you test if Facebook incorporating colored backgrounds to statuses improves user engagement?

1. Clarify goal of the feature ?

- Is this feature for specific user group (or) for all user groups?
- what do you hope to acheive with this feature incorporatin
- why this specific feature update for that goal and not any other feature ?

Made with Goodnotes

=> Design metrics:

1. User Engagement : % of active users who have engaged with fb (comments, likes, save, reactions --->)

2. Daily Active users: # of unique users who have logged on fb each day

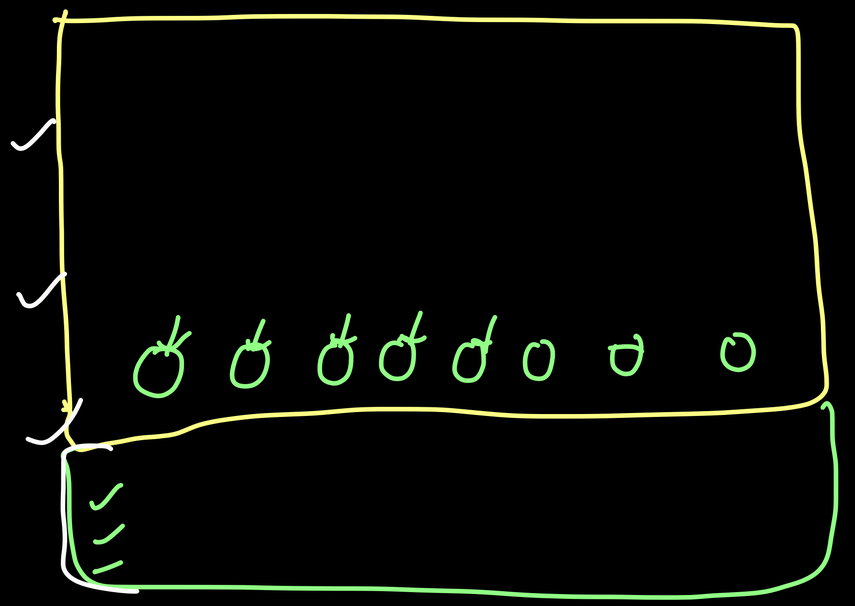
=> North star Metric. % of user with Engagement

=> Supporting metric (LI) : DAU [Daily active users]

Made with Goodnotes

→ Gaurd rail metric :

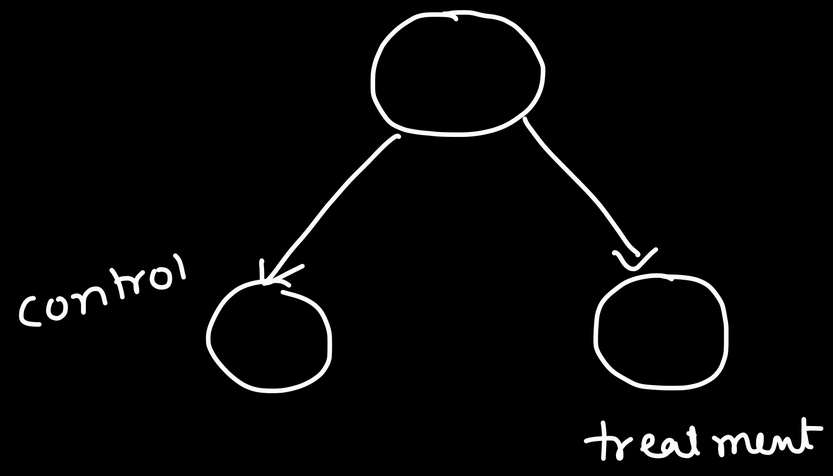
- % Media content



** Experimentation

1. Set up Hypothesis :

- H_0
- H_a



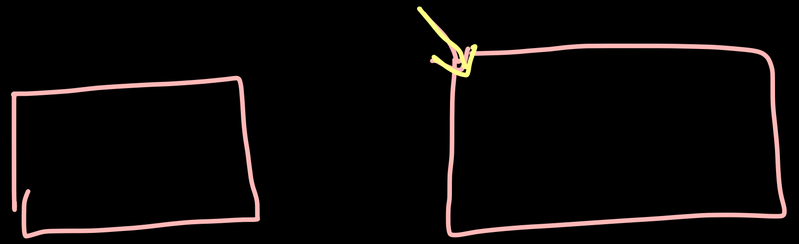
H_0 : No significant difference in user engagement b/w control & treatment group.

H_a : There is a significant difference in user Engagement b/w control & treatment group.

Made with Goodnotes

2. Choice test : z- propotion test ⇒

3. Sample size calculation



(1) Base line metrics:

assumption 45%

(2) Minimum detectable effect :

13 → 0.1%

10 B

45%

10 M

45%

45.1%

0.1%

What change is considered meaning-ful enough for you to take an action

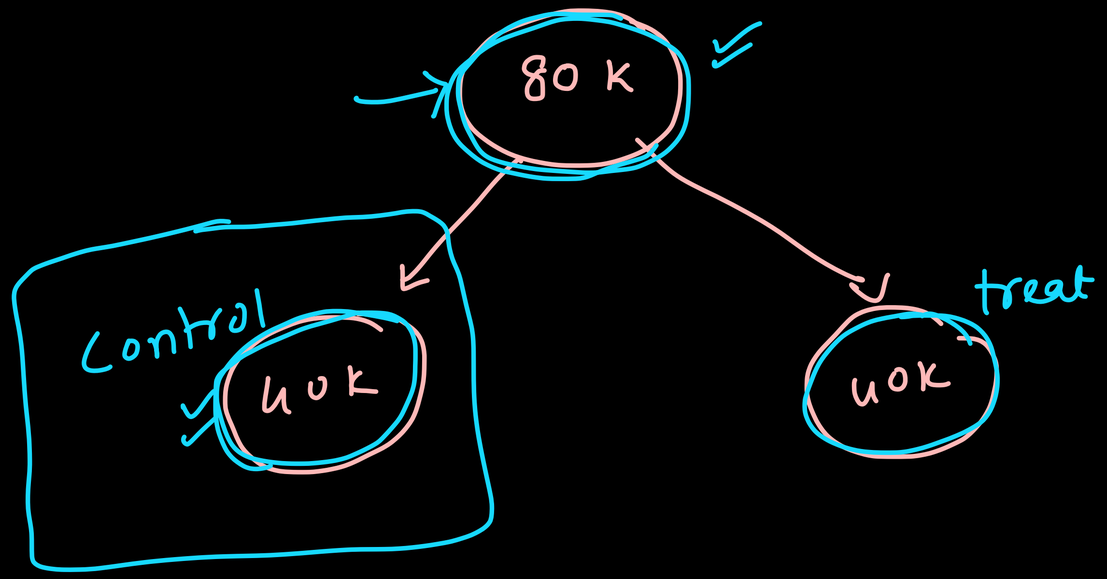


→ Confidence level [95%]

→ Power [80%]

↓

(1-β)



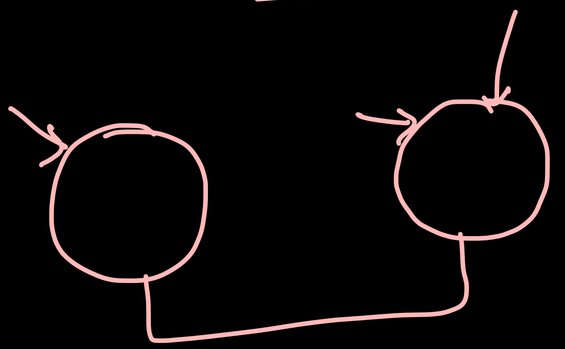
Made with Goodnotes

→ Experiment Duration

1.

FB → 5k | day

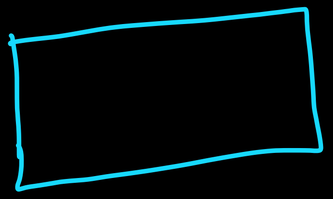
A | B



Duration => 90k / 5k = 16 days

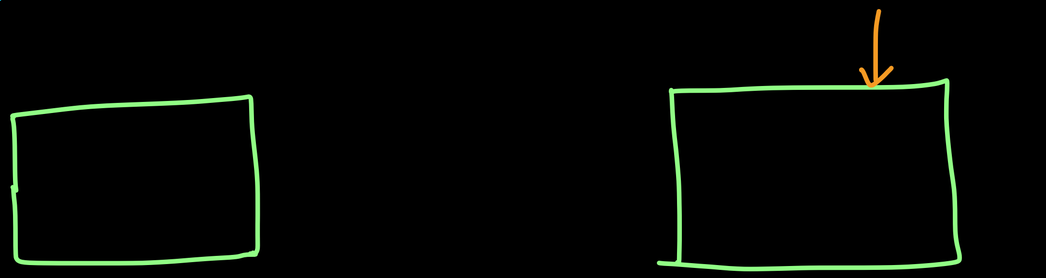
Testing pitfalls

1. primary Effect & Novelty Effect.



Made with Goodnotes

1. Primary Effect :



control

treatment

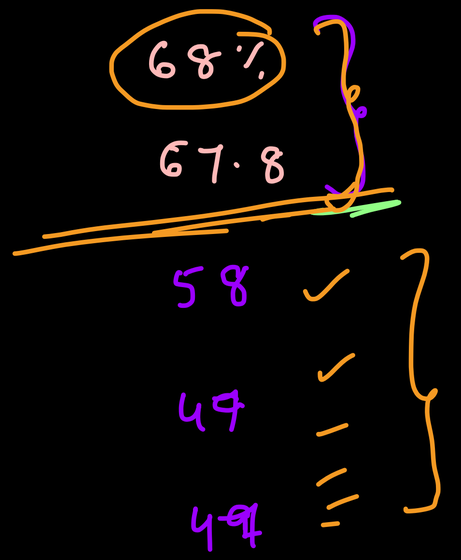


	control	treatment
week 1	45 %	45.001
		45.525
week 2	45.5 %	
	46 %	49
	47 %	57 %
	⋮	

2. Novelty Effect

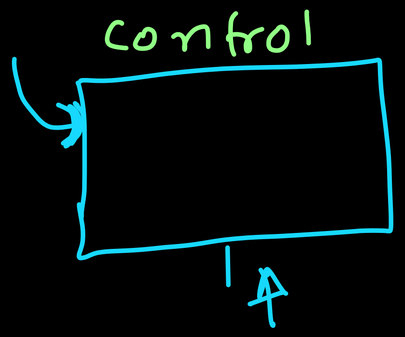
Instagram Thread

	Control	treatment
week 1	45%	68%
week 2	45.5%	67.8
⋮	47%	58 ✓
		49 ✓
		49 =

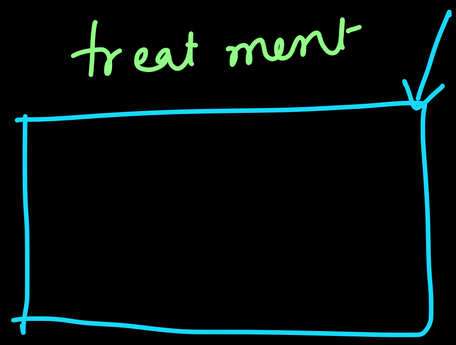


⇒ Group Interference

✓ Network effect :



⇒ 70%



Youtube influencer

Recommendations based on experiment results — Launch (or) not?

1.

1%. ⟶ $20 M ⇒ 

2. Is it worth it to launch the product given all costs?

⇒ 45%. ↳ 55%. % user Engagement

↓ DAU

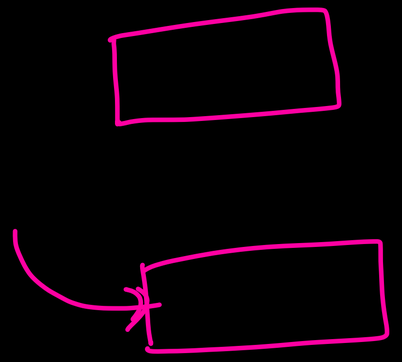
1M 90k

An e-commerce website wants to optimize its checkout page. They run an A/B test where half of the visitors see a green "Buy Now" button, while the other half sees a red "Buy Now" button. What could be the goal of this experiment? Choose the most appropriate answer.

11 users have participated

A	The goal is to determine which color increases the click-through rate and leads to more purchases.	45%
B	The goal is to assess whether changing the button color impacts customer satisfaction with the website design.	55%
C	The goal is to analyze the impact of button color on the average time spent on the checkout page.	0%

End Quiz Now



A media company is testing different subject lines for its weekly newsletter. They send two versions of the newsletter, each with a different subject line, to a subset of their subscribers. What would be the most appropriate action item for the media company as per the ongoing experiment?

2 users have participated

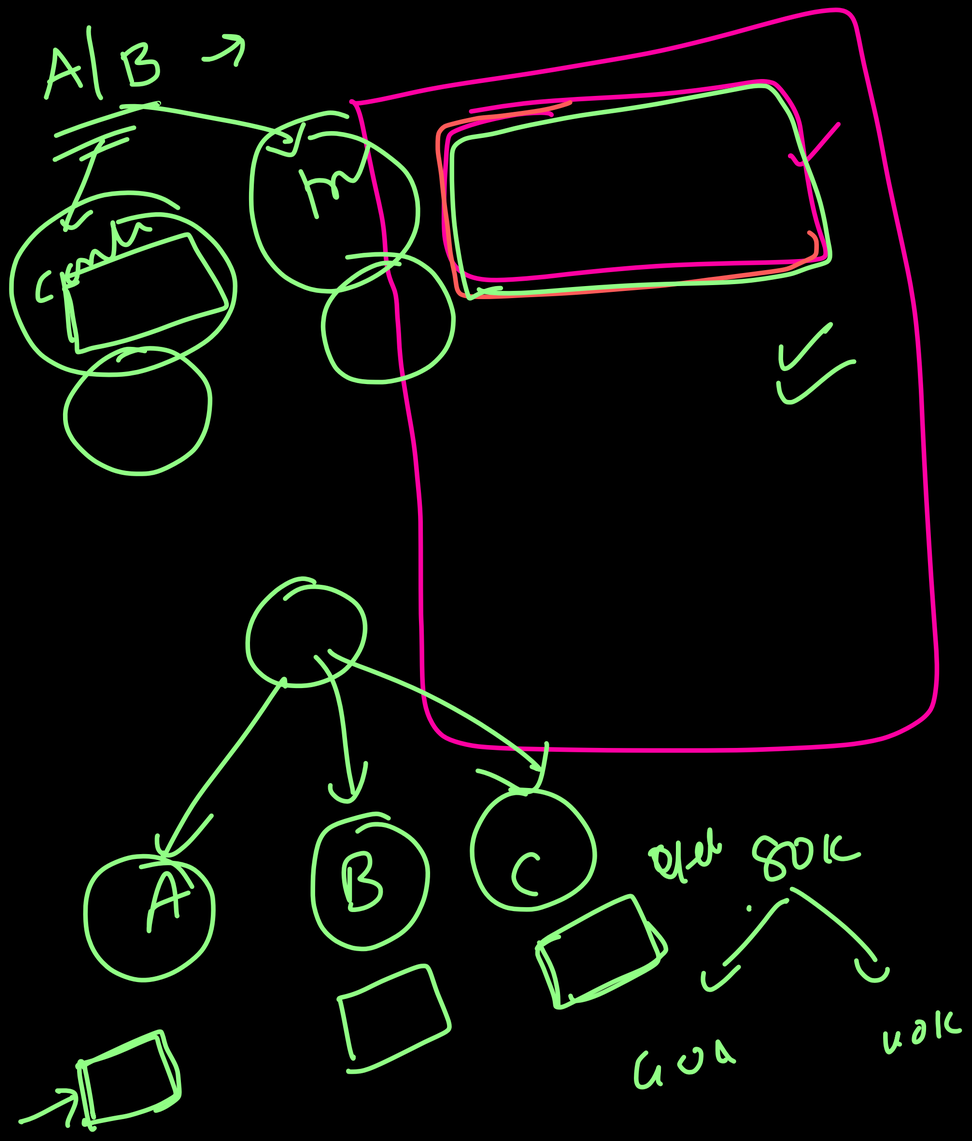
A	They conduct a survey to gather feedback from subscribers about the subject line for their respective group.	0%
B	They change the font style & color among the two versions of the newsletter to see if it has any impact on subscriber interaction.	50%
C	They increase the frequency of newsletter distribution for one group to see if more frequent newsletters result in higher engagement.	0%
✓ D	They measure open rates to determine which subject line is more effective at capturing reader attention.	50%

Made with Goodnotes

An online retailer is experimenting with the layout of its product pages. They test two variations: one with the product description at the top and one with customer reviews at the top. What would be the most appropriate choice of next step in the currently ongoing experiment?

3 users have participated

A	They analyze the number of social media shares for each product to understand the impact of the layout variations on brand popularity.	67%
✓ B	The company measures the click-through rate and conversion rate to determine which layout leads to more sales.	0%
C	They conduct a survey to gather customer preferences on the more appealing layout of the product pages of the website.	33%



Made with Goodnotes